Xiaochan Xue, Ph.D.

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Research Profile

I am a **tenure-track Assistant Professor** in the Department of Electrical and Computer Engineering at the University of Hawai'i at Mānoa. My research brings together **wireless communications, cybersecurity, and artificial intelligence**, with a long-term vision of enabling **trustworthy physical AI over next-generation wireless networks**.

My group develops secure and resilient AI-native wireless systems through physical-layer security, integrated sensing and communication (ISAC), and learning at the radio access network edge. Current directions include AI-RAN/O-RAN security and prototyping, wireless sensing, distributed learning, edge intelligence, and intelligent agents for adaptive, context-aware decision making.

Academic Appointments

08/2025 – Present — **Assistant Professor (Tenure Track).** Department of Electrical and Computer Engineering, College of Engineering, University of Hawai'i at Mānoa.

2024 – 2025 — **Lecturer.** Department of Electrical and Computer Engineering, Stevens Institute of Technology.

Education

2020 – 2025 — **Ph.D., Computer Engineering,** Stevens Institute of Technology
Department of Electrical and Computer Engineering

2018 – 2019 — **M.S., Electrical Engineering,** Stevens Institute of Technology
Department of Electrical and Computer Engineering

2013 – 2017 — **B.E., Communication Engineering,** Jilin University
Department of Communication Engineering

Research Funding

06/2026 — **NVIDIA Academic Grant Program**, NVIDIA Corporation. **PI**.
Four NVIDIA RTX PRO 6000 Blackwell GPUs, approximately **$70,000** in value; received in August 2026. Supports AI-native wireless systems, AI-RAN, cybersecurity, edge AI, and semantic ISAC research.

Genesis Mission Phase I, U.S. Department of Energy. **Co-PI**.
Total award: **$750,000**; University of Hawai'i allocation: **$420,000**.

Honors and Awards

10/2026 — **Outstanding Service Award**, IEEE Healthcom 2026. Co-organized WCIHS 2026 and the poster session with Yucheng Xie.

2025 — **Ph.D. Fellow**.

Honors and Awards (continued)

02/2025 **NSF CPS Rising Star**, Vanderbilt University, Nashville, TN.

2023 – 2024 **Excellence Doctoral Fellowship**, Stevens Institute of Technology.

Graduate Conference Fund, Stevens Institute of Technology.

2019 – 2020 **Research Assistantship**, Stevens Institute of Technology (master's studies).

10/2016 **Third Class Scholarship**, Jilin University.

Outstanding Student Leader, Jilin University.

10/2015 **Excellent Student Award in College**, Jilin University.

Research Publications

[#] Corresponding author. Accepted papers are marked explicitly.

Conference Proceedings

1. **X. Xue**, S. Yu, S. Parkar, and J. Li, "Waveform design for ISAC: Trends and future directions," in *Proceedings of the IEEE International Conference on Computing, Networking and Communications (ICNC)*, 2026. URL: https://ieeexplore.ieee.org/abstract/document/11416964.

2. **X. Xue**[#], Z. Zhang, and S. Yu, "TADD: Topology-aware distributed distillation in AI-native 6G RAN," in *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*, Accepted, Macau, China, 2026. URL: https://luna-xue.github.io/assets/files/2026_GLOBECOM_TADD.pdf.

3. C.-H. Yu, T. Yang, S. Parkar, H. Cai, Y. Pan, S. Wu, **X. Xue**, and Y. Zheng, "Programmable transmitarray-assisted near-field ISAC for in-cabin occupancy sensing," in *2026 IEEE 1st Annual Integrated Sensing and Communication Conference (ISAC)*, Accepted, Lisbon, Portugal, 2026.

4. **X. Xue**, S. Parkar, S. Yu, and Y. Zheng, "AI-assisted composite ISAC for mmWave respiration pattern recognition," in *Proceedings of the IEEE Annual Congress on Artificial Intelligence of Things (AIoT)*, 2025. URL: https://ieeexplore.ieee.org/abstract/document/11416194.

5. **X. Xue**, S. Yu, S. Parkar, and Y. Zheng, "ROISD: RIS and O-RAN assisted intelligent sensing for UAV detection," in *Proceedings of the IEEE Annual Congress on Artificial Intelligence of Things (AIoT)*, 2025. URL: https://par.nsf.gov/servlets/purl/10662511.

6. **X. Xue**, M. K. Hasan, S. Yu, L. N. Kandel, and M. Song, "Over-the-air federated learning with enhanced privacy," in *ICC 2023 - IEEE International Conference on Communications*, 2023, pp. 4546–4551. DOI: 10.1109/ICC45041.2023.10278765.

7. **X. Xue**, S. Yu, and M. Song, "Secure device trust bootstrapping against collaborative signal modification attacks," in *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications*, 2023, pp. 1–10. DOI: 10.1109/INFOCOM53939.2023.10229007.

8. **X. Xue**, S. Yu, M. Song, and C. Xin, "K-group random channel hopping (K-RCH) rendezvous for cognitive radio networks," in *ICC 2021 - IEEE International Conference on Communications*, 2021, pp. 1–6. DOI: 10.1109/ICC42927.2021.9500643.

9. H. Wu, Y. Zou, J. Zhu, **X. Xue**, and T. Tsiftsis, "Secrecy performance of hybrid satellite-terrestrial relay systems with hardware impairments," in *ICC 2019 - 2019 IEEE International Conference on Communications (ICC)*, 2019, pp. 1–6. DOI: 10.1109/ICC.2019.8761231.

Journal Articles

1. M. K. Hasan, **X. Xue**, S. Yu, and M. Song, "Cooperative NOMA-based spectrum leasing with multiple secondary users," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 11, pp. 14 543–14 558, 2023. DOI: 10.1109/TVT.2023.3287230.

Workshop and Poster Publications

1. R. Xu, Y. Wan, and **X. Xue**#, "DomusMind: A benchmark for evaluating lifelong smart home agents under drift," in *ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving*, 2026. URL: https://openreview.net/forum?id=dCBF23RZYJ.

Preprints and Manuscripts Under Review

1. S. Parkar, **X. Xue**#, S. Li, and Y. Zheng, "SPECTRA: Spectrally disjoint OFDM–FMCW mmWave ISAC with utility-driven mode selection," Preprint; submitted to IoT-J; under review, 2026. URL: https://luna-xue.github.io/assets/files/2026_IOT_J_SPECTRA.pdf.

Prior Research Experience

01/2022 – 08/2025

- **Research Assistant.** Stevens Institute of Technology, AISecLab
  Advisor: Prof. Shucheng Yu
- **Physical-Layer Security for Distributed Wireless Devices**
  – Developed a novel trust bootstrapping protocol to mitigate collaborative RF signal modification attacks without additional hardware.
  – Designed lightweight, scalable PLS-based confidential computing techniques for federated learning (FL) and multi-device collaboration, including a pairwise cancellable random artificial noise (PCR-AN) scheme for AirComp-based FL.
  – Conducted theoretical analysis and experimental validation to ensure security, privacy, and FL convergence.
- **mmWave Sensing and Integrated Sensing and Communication (ISAC)**
  – Built an AI-driven sensing system for contactless breathing pattern detection using mmWave FMCW radar, achieving high real-world detection accuracy.
  – Implemented OFDM-based respiration sensing and explored neural network integration for data processing and pattern recognition.
- **Wireless Sensing in O-RAN with Security and AI**
  – Designed wireless sensing applications for distributed learning under O-RAN.
  – Explored generative AI-integrated ISAC in AI-RAN/O-RAN to enhance cyber-physical systems.

05/2019 – 12/2019

- **Research Assistant.** Stevens Institute of Technology
  Advisor: Prof. Min Song
  Spectrum (Channel) Utilization Optimization
  – Designed and implemented K-RCH, a group-based random channel hopping protocol for rendezvous in Cognitive Radio Networks.
  – Improved spectral efficiency, reduced time-to-rendezvous (TTR), and enhanced throughput in dynamic spectrum environments.

Prior Research Experience (continued)

06/2018 – 08/2018

Visiting Researcher. Nanjing University of Posts and Telecommunications
Advisor: Prof. Yulong Zou
Secrecy Performance of Hybrid Satellite-Terrestrial Relay Systems
– Analyzed secrecy outage probability under hardware impairments (HIs) in hybrid satellite-terrestrial relay systems.
– Derived closed-form results and investigated impacts of relay location, eavesdropper position, and transceiver impairments on secure communication performance.

Teaching Experience

Fall 2026

Instructor. University of Hawai'i at Mānoa
Course: ECE 362: Discrete Math for Engineers.

Instructor. University of Hawai'i at Mānoa.
ENGR 296: Sophomore Vertically Integrated Project (VIP).
Project-based research on robotic car control, physical AI, and O-RAN-assisted experimentation.

Spring 2026

Instructor. University of Hawai'i at Mānoa
Course: ECE 669: Wireless and Mobile Security.

Fall 2025

Instructor. University of Hawai'i at Mānoa
Course: ECE 693E: Intelligent and Secure Wireless Networks.
From Wireless Fundamentals to intelligent, secure, and programmable AI-driven Radio Access Networks.

2024 – 2025

Instructor. Stevens Institute of Technology
Course: CPE 691: Information Systems Security (Graduate Level, On-site & WebCampus).
Designed and delivered course content covering cybersecurity fundamentals, network security, cryptography, and system vulnerabilities.
Developed hands-on assignments and projects to help students understand secure communication protocols, authentication mechanisms, and emerging threats.

2019 – 2022

Teaching Assistant. Stevens Institute of Technology
Course: Applied Machine Learning (Graduate Level).
Supported course instruction by designing assignments, mentoring students during office hours, and guiding them through supervised/unsupervised learning and neural network techniques.
Course: Engineering Design IV (Undergraduate Level).
Instructed this senior design laboratory course, mentoring undergraduates in developing design skills and engineering judgment through design projects in communication, control systems, signal processing, and MATLAB-based simulations.

2018 – 2019

Grader. Stevens Institute of Technology (during master's studies)
Course: Applied Machine Learning (Graduate Level).
Course: Linear System Theory (Graduate Level).

Student Advising and Mentoring

Ph.D. Advisor. *Department of Electrical and Computer Engineering, UHM*

○ **Saurabh Parkar** *2026–Present*

Advisor for research on O-RAN, AI-native wireless systems, integrated sensing and communications (ISAC), and wireless cybersecurity. Recipient of the UHM Giving Tree Scholarship for the 2026–2027 academic year.

Master's Research Mentoring. *WINGS Lab, University of Hawai'i at Mānoa*

○ **Aris Carlos** *Current*

O-RAN/AI-RAN infrastructure and cellular security / fake base station research.

○ **Ethan Morrell** *Current*

O-RAN/AI-RAN infrastructure and system fuzzing.

○ **Thomas Yang; Matthew Matsuo** *Alumni*

Former master's students in WINGS Lab.

Ph.D. Committee. *Department of Electrical and Computer Engineering, University of Victoria*

○ **Hyemin Yu** *Summer 2026*

Defense External Examiner.

Ph.D. Committee. *Department of Electrical and Computer Engineering, UHM*

○ **Cho-hao Yu** *Summer 2026*

Qualifying Examination Committee.

○ **Ziyi Zhuang** *Fall 2025*

Qualifying Examination Committee.

Undergraduate Research Mentor. *University of Hawai'i at Mānoa*

○ **Tomohiro Yamada (Boston University)** *Summer 2026*

Mentoring on edge AI systems, robotics operation, system logging, and wireless/network performance documentation.

VIP Project. *Department of Electrical and Computer Engineering, UHM*

○ **Jingkai Ni; John Kirby Ramirez; Josiah Farias** *Fall 2026*

Undergraduate VIP project and AFRL SDR Challenge 2026.

○ **Trevor Suzuki** *Fall 2026*

Undergraduate VIP project.

JESSE Summer Intern Mentor. *Department of Electrical and Computer Engineering, UHM*

○ **Kensei Narahara** *2026*

Assisted with O-RAN testbed deployment, software installation, experiment documentation, and system validation.

○ **Tim He** *2026*

Participated in wireless sensing, AI-enabled networking experiments, and performance evaluation.

○ **Jayden Lau** *2026*

Supported data collection, laboratory operation, and dissemination of research outcomes.

Graduate Mentor. *Department of Electrical and Computer Engineering, Stevens Institute of Technology*

Supervised and mentored M.S. students in Professor Shucheng Yu's research group.

○ **Jiapeng Xiao** *Spring 2021*

Guided a one-semester master's research scholarship; designed projects and trained him in USRP SDR operation for hardware-software integration.

○ **Zhifan Jiang** *2022–2023*

Designed a one-year project on LabVIEW and TMYTEK mmWave antennas; advised research idea development and report writing. Recipient of the *Excellent Research Award*.

○ **Beula Jose** *2023*

Directed a one-semester project on adversary experiments using USRP SDR; enhanced research skills and project performance.

○ **Ruoxi Li** *2024–2025*

Introduced communication systems research using TMYTEK mmWave antennas and OFDM transmission in LabVIEW; advised CSI-based breathing signal collection and data analysis. Subsequently secured a Ph.D. offer.

○ **Saurabh Parkar** *2024–2025*

Supervised projects on O-RAN with POWDER and srsRAN using USRP SDR; trained for presentations and technical reporting. Recipient of the *1st-rank Department Scholarship*. Co-developed projects on mmWave FMCW sensing and 5G NR OFDM respiration monitoring.

Team Mentor. *AFRL Software Defined Radio (SDR) Challenge*

○ **AFRL SDR Challenge Team** *2024–2025*

Mentored undergraduate and high-school students in developing wireless communication projects using USRP SDR platforms, including MIMO implementation in MATLAB. Undergraduate researchers included Lenny Yanza, Karen Caguana, and Shaun George.

○ **AFRL SDR Challenge Team** *2025–2026*

Led a collaborative team involving Stevens Institute of Technology and Yeshiva University on a RIS-assisted drone detection project using software-defined radio platforms.

Professional Service

Leadership and Conference Organization

2027 **Local Organizing Chair**, IEEE INFOCOM 2027 (appointed 2026).

2026 **Geo-diversity Co-chair**, N²Women Board.

WCIHS Workshop Chair, IEEE Healthcom 2026.

Demo/Posters Chair, IEEE Healthcom 2026.

Session Chair, IEEE ICNC 2026.

Technical Program Committees

2027 IEEE ICCCN; IEEE INFOCOM; IEEE Symposium on Security and Privacy (S&P).

2026 International Performance, Computing and Communications Conference (IPCCC).

The 2nd free5GC World Forum.

IEEE/ACM CHASE, mHealth Security Workshop.

IEEE Healthcom.

Professional Service (continued)

- IEEE INFOCOM (Main Track); IEEE INFOCOM ISAC-FutureG Workshop / ISAC Track.

Federal and External Review Service

- **Panelist**, U.S. National Science Foundation (NSF), CyberAICorps Scholarship for Service (CyberAI SFS).

Peer Review

2024 – Present

- **Journals:** IEEE Transactions on Wireless Communications (TWC); IEEE Transactions on Vehicular Technology (TVT); IEEE Internet of Things Journal (IoT-J); AJECE.
  Conferences and workshops: IEEE S&P 2027; NeurIPS 2026; IEEE INFOCOM 2025–2026; IEEE/ACM CHASE 2026; IEEE Healthcom 2026; IEEE CNS 2026; ICLR 2026 AIWILD; ICML 2026 AIWILD.

Institutional Service

2025 – 2026

- **Research Committee Member, UHM, US**

2026

- **Publicity Committee, Electrical and Computer Engineering, UHM, US**

Professional Memberships and Lab Affiliations

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE).
- IEEE Communications Society.
- IEEE Women in Engineering.
- IEEE Young Professionals.
- Association for Computing Machinery (ACM).

WINGS Lab Organizational Affiliations

08/2026

- OpenAirInterface, member organization.

06/2026

- OCUDU and Linux Foundation, associate membership.

Presentations and Seminars

Presentations

04/2025

- X. Xue, S. Yu, *"AI-Driven Integrated Sensing and Communication (ISAC) in AI-RAN/O-RAN: Scalability, Privacy, and Security"*, Stevens Institute of Technology ECE Department Research Expo, NJ, USA. (Poster presentation)

02/2025

- X. Xue, S. Yu, *"AI-Driven Integrated Sensing and Communication (ISAC) in AI-RAN/O-RAN: Scalability, Privacy, and Security"*, NSF CPS Rising Star Workshop, Nashville, TN, USA. (Poster, 17% acceptance rate)

Presentations and Seminars (continued)

08/2024 X. Xue, S. Yu and M. Song, *"Secure Device Trust Bootstrapping Against Collaborative Signal Modification Attacks"*, 1st Symposium on Emerging Topics in Networks, Systems, and Cybersecurity, Stevens Institute of Technology, NJ, USA. (Poster presentation)

05/2023 X. Xue, M. K. Hasan, S. Yu, L. N. Kandel, and M. Song, *"Over-the-Air Federated Learning with Enhanced Privacy"*, Rome, Italy. (Invited oral presentation on conference paper)

X. Xue, S. Yu and M. Song, *"Secure Device Trust Bootstrapping Against Collaborative Signal Modification Attacks"*, New York area, USA. (Invited oral presentation on conference paper)

06/2021 X. Xue, S. Yu, M. Song and C. Xin, *"K-Group Random Channel Hopping (K-RCH) Rendezvous for Cognitive Radio Networks"*, virtual due to COVID. (Invited oral presentation on conference paper)

Invited Seminars

09/2026 X. Xue, *"AI-RAN for Intelligent Wireless Sensing: From Waveform Design to Task-Aware Scheduling"*, University of Hawai'i at Mānoa, HI, USA. (Invited oral seminar)

12/2025 X. Xue, *"Integrated Sensing and Communication in FutureG Systems"*, Yeshiva University, NY, USA. (Invited oral seminar)

02/2025 X. Xue, *"Enhancing Security and Privacy in Distributed Wireless Networks Through Physical Layer Techniques"*, Stevens Institute of Technology, NJ, USA. (Invited oral seminar)

Professional Development

06/2026 **GW AI in Cybersecurity Education Faculty Development Institute**, George Washington University.

12/2025 **NSF/SaTC Aspiring PI Workshop**, George Washington University, Washington, D.C. (Invited participant.)

Collaborators

Dr. Shucheng Yu, Yeshiva University, USA

Dr. Min Song, Stevens Institute of Technology, USA

Dr. Laxima Niure Kandel, Embry-Riddle Aeronautical University, USA

Dr. Chunsheng Xin, Old Dominion University, USA

Dr. Yao Zheng, University of Hawai'i at Mānoa, USA

Dr. Hanqing Guo, University of Hawai'i at Mānoa, USA

Dr. Shuang Li, University of Victoria, Canada

Dr. Changzhi Li, Texas Tech University, USA

Dr. Yinxin Wan, UMass Boston, USA

Dr. Yulong Zou, Nanjing University of Posts and Telecommunications, China

Skills

Languages	Strong reading, writing and speaking competencies for English, Mandarin Chinese.
Coding	MATLAB, Python, LabVIEW, GNU Radio, Visual Studio, C/C++, Java, R
Web Dev	HTML, CSS, JavaScript.
Platform	POWDER, COSMOS, open-source srsRAN
Hardware	Software-Defined Radios (SDRs): USRP B/N series, NI USRP 2974 Antennas: mmWave (beamforming), low-frequency (directional, omni-directional) Measurement equipment: Keysight, Rohde & Schwarz
Others	Unity, Final Cut Pro, Studio One, OBS

Relevant Coursework

Wireless & Security	Wireless Communications; Communication Theory; Information Systems Security; Computer Network; Information & Coding Theory.
Machine Learning & Computing	Applied Machine Learning; Algorithms & Data Structures; Computing Principles for Embedded Systems; Computing Principles for Mobile and Embedded Systems; Computer Architecture; Python; C++.
Mathematics & Theory	Advanced Mathematics; Probability Theory & Statistics; Complex Analysis & Integral Transformation; Analytical Methods in Electrical Engineering; Random Signal Analysis.
Circuits & Signal Processing	Digital Signal Processing; Signals & Systems; Data Acquisition; Digital Circuit & Logic Design; Analog Electronic Circuits; High-Speed Circuit Signal Integrity Analysis & Design.
Control	Introduction to Control Theory.